

AI Technical Program Manager – 60 Day Learning Roadmap

Companion Workbook to the AI Technical Program Manager Handbook

Target Audience

- QA Leads
 - Delivery Managers
 - Technical Program Managers
 - Engineering Managers
 - Product Managers
 - AI TPM Aspirants
-

Primary Goal

By Day 60, learners will be able to:

1. Explain AI concepts confidently.
 2. Lead enterprise AI initiatives.
 3. Understand the Microsoft Azure AI ecosystem.
 4. Design enterprise AI architectures.
 5. Build an AI TPM portfolio.
 6. Successfully navigate AI TPM interviews.
 7. Transition into AI TPM roles.
-

Cover

AI Technical Program Manager

60-Day Learning Roadmap

From QA Lead to AI Leadership

Companion Workbook Includes:

- 8 Weekly Learning Tracks
 - 60 Daily Lessons
 - Microsoft Learn References
 - Azure Hands-On Labs
 - AI TPM Templates
 - Prompt Libraries
 - Interview Preparation
 - Capstone Project
-

Weekly Themes

Week	Theme	Deliverable
Week 1	AI Foundations	Explain AI to CEO, CTO, Developer
Week 2	Azure AI Ecosystem	Azure AI Architecture Diagram
Week 3	Prompt Engineering	TPM Prompt Library
Week 4	LLMs & RAG	Migration Documentation Chatbot
Week 5	AI Agents	TPM Agent Suite
Week 6	AI Product Management	Product Charter & Cost Sheet
Week 7	System Design	Enterprise AI Assistant
Week 8	Career Preparation	Resume, Portfolio & LinkedIn
Day 60	Capstone	AI TPM Portfolio Package

WEEK 1

Theme: AI Foundations

"The future belongs to leaders who understand both technology and business."

Weekly Objectives

- Understand AI, ML, DL, and GenAI.
- Learn tokens, embeddings, and context windows.
- Build confidence explaining AI concepts.
- Complete AI Foundations chapters (1–3).

Weekly Milestone

Deliver a 5-minute presentation explaining AI to:

- CEO
- CTO
- Software Developer

Day 1

Field	Details
Learning Objective	Understand AI fundamentals

Field	Details
Handbook Reference	Chapter 1
Concepts	AI, ML, DL
Microsoft Learn	https://learn.microsoft.com/training/paths/get-started-with-artificial-intelligence-on-azure/
Free Tool	ChatGPT Free
Hands-On	Create AI mind map
Deliverable	AI One-Pager
Time	60 Minutes
Reflection	"What business problems can AI solve?"

Day 2

Field	Details
Objective	AI vs ML vs DL
Concepts	Enterprise adoption
Tool	Gemini
Activity	Compare use cases
Deliverable	Comparison Table
Reflection	"Where does my organization sit on the AI maturity curve?"

Day 3

- Generative AI
- Enterprise Use Cases
- AI Adoption

Deliverable: Top 10 GenAI use cases for your company.

Day 4

- Large Language Models
- GPT
- Gemini
- Claude
- Phi

Deliverable: Create an LLM comparison matrix.

Day 5

- Tokens
- Context Windows
- Temperature
- Hallucinations

Prompt:

"Explain tokens and context windows to a non-technical executive using airline examples."

Day 6

- Embeddings
- Semantic Search
- Vector Concepts

Hands-On

- Use NotebookLM.
 - Upload a PDF.
 - Observe semantic retrieval.
-

Day 7

Weekly Review

Checklist

- ☐ Completed Chapter 1
 - ☐ Completed Chapter 2
 - ☐ Completed Chapter 3
 - ☐ Created AI One-Pager
 - ☐ Presented AI concepts
-

WEEK 2

Theme: Azure AI Ecosystem

"Cloud is no longer an advantage. Knowing how to leverage it is."

Topics

- Azure OpenAI
- Azure AI Foundry
- Azure AI Search
- Azure Functions
- Azure Blob Storage
- Azure Monitor

Microsoft Learn

- Azure OpenAI Fundamentals
- Azure AI Foundry
- Azure AI Search

Deliverable

Create an Azure Architecture Diagram.

Days 8-14

Day	Topic	Deliverable
8	Azure OpenAI	Service Overview
9	Azure AI Foundry	Foundry Notes
10	Azure AI Search	Search Architecture
11	Azure Functions	Workflow Diagram
12	Azure Blob Storage	Storage Design
13	Azure Monitor	Monitoring Dashboard
14	Weekly Review	Azure Architecture Diagram

WEEK 3

Theme: Prompt Engineering

"Prompting is the new management skill of the AI era."

Topics

- Zero Shot
- Few Shot
- Chain of Thought
- Structured Outputs
- Function Calling

Deliverables

Create:

1. Sprint Planning Prompt
2. Risk Register Prompt
3. Executive Summary Prompt

Example Prompt

"Act as a Senior AI TPM. Create a RAID log for a Teradata to Azure migration involving 12 teams and a 6-month timeline."

WEEK 4

Theme: LLMs & RAG

Topics

- GPT
- Claude
- Gemini
- Phi
- RAG
- Vector Databases
- Chunking

Deliverable

Design a Migration Documentation Chatbot.

Suggested Tools

- NotebookLM
- Open WebUI
- Ollama
- Langflow

Suggested GitHub Repositories

- LangChain
 - CrewAI
 - Flowise
 - Open WebUI
-

WEEK 5

Theme: AI Agents

Topics

- Tool Calling
- Agent Memory
- Planning
- Multi-Agent Systems

Deliverables

Build:

- Meeting Minutes Agent
- RAID Agent
- Sprint Planning Agent

Suggested Tools

- CrewAI
 - LangGraph
 - Flowise
 - Azure AI Foundry Agents
-

WEEK 6

Theme: AI Product Management

Topics

- Product Lifecycle
- Governance
- Responsible AI
- Evaluation
- Cost Estimation

Deliverables

- AI Product Charter
 - Cost Estimation Worksheet
 - Governance Checklist
-

WEEK 7

Theme: System Design

Topics

- APIs
- Authentication
- Monitoring
- CI/CD
- Enterprise Architecture

Deliverable

Design an Enterprise AI Assistant.

Architecture Components

```
graph TD
  A[User]
  --> B[Frontend]
  B--> C[API Gateway]
  C--> D[Azure OpenAI]
  D--> E[Azure AI Search]
  E--> F[Database]
  F--> G[Monitoring]
```

WEEK 8

Theme: AI TPM Career Preparation

"Your career compounds in the same way your skills do."

Topics

- STAR Stories
- Interview Preparation
- Resume
- LinkedIn
- Leadership
- Stakeholder Management

Deliverables

- AI TPM Resume
- LinkedIn About Section

- Portfolio Website
- Interview Cheat Sheet

Top Interview Questions

1. Explain RAG.
2. What are AI Agents?
3. How do you manage risks?
4. Describe a migration program.
5. How would you build an enterprise AI assistant?
6. Explain Responsible AI.
7. How do you estimate AI costs?
8. Describe a difficult stakeholder.
9. Tell me about a failure.
10. Why AI TPM?

DAY 60 CAPSTONE

Build Your AI TPM Portfolio

Required Deliverables

Deliverable	Status
AI TPM Portfolio	
Azure Architecture Diagram	
RAID Log	
Sprint Plan	
AI Governance Checklist	
Product Charter	
Executive Dashboard	
AI Assistant Architecture	
Resume	
LinkedIn About Section	

Progress Tracker

Week	Complete
Week 1	☐
Week 2	☐

Week	Complete
Week 3	☐
Week 4	☐
Week 5	☐
Week 6	☐
Week 7	☐
Week 8	☐
Capstone	☐

Recommended Free Tools

Category	Tool
AI Assistant	ChatGPT
Research	Gemini
Documents	NotebookLM
Agents	CrewAI
Workflow	Flowise
Local Models	Ollama
Diagrams	Mermaid
Project Management	Trello
Portfolio	GitHub Pages

Recommended GitHub Repositories

- LangChain
 - CrewAI
 - LangGraph
 - Flowise
 - Open WebUI
 - Microsoft Semantic Kernel
 - AutoGen
 - Ollama
 - Continue.dev
 - OpenHands
-

Final Message

By completing this roadmap, you will possess:

- AI Fundamentals
- Azure AI Knowledge
- System Design Skills
- Program Management Expertise
- Leadership Stories
- Interview Readiness
- A Complete AI TPM Portfolio

You will not merely understand AI—you will be prepared to lead enterprise AI programs.

****charan.ai | ****

© 2026 Charan.ai. All Rights Reserved.